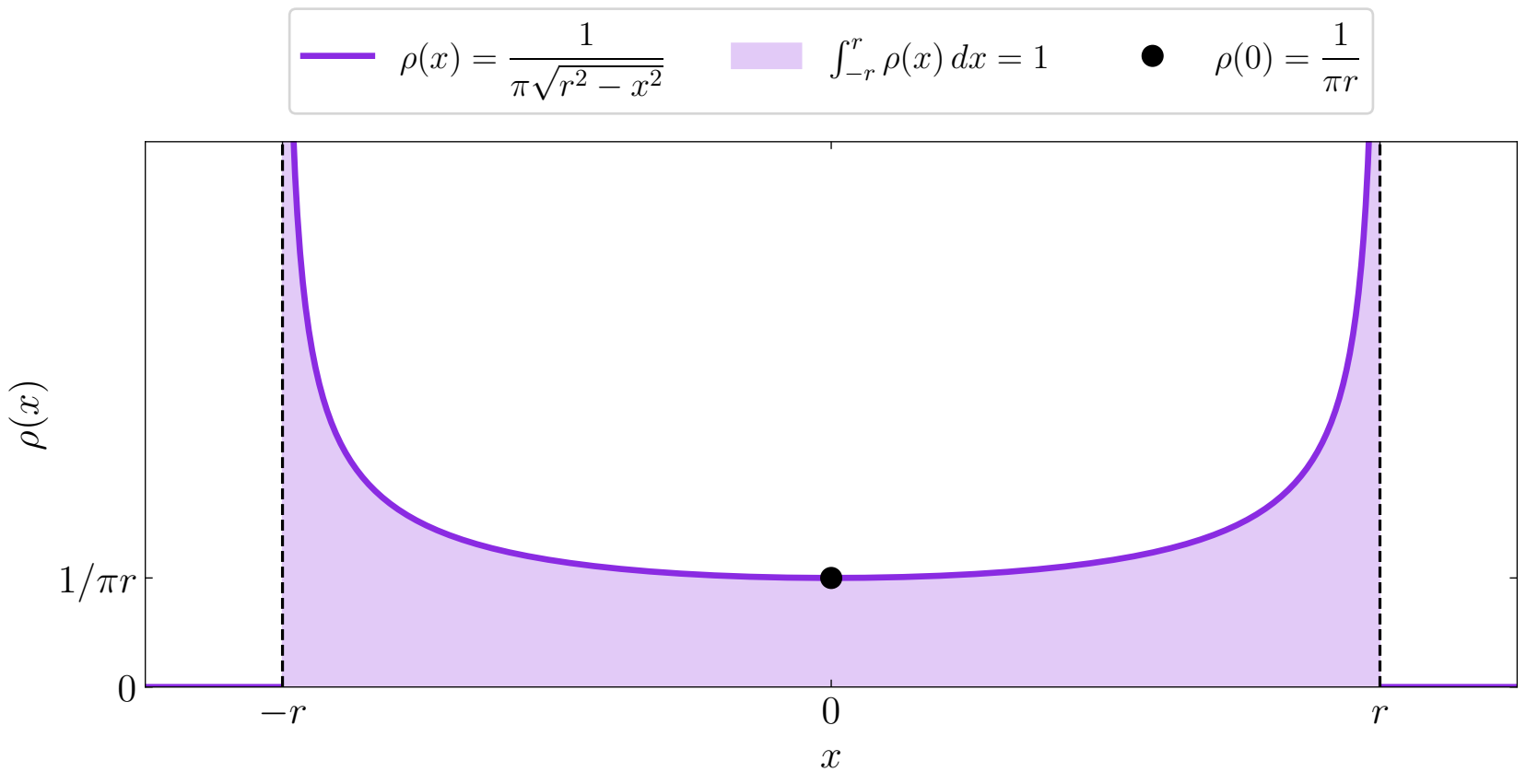

$\rho(x) = \frac{1}{\pi\sqrt{r^2 - x^2}}$
$\int_{-r}^{r} \rho(x)\, dx = 1$
$\rho(0) = \frac{1}{\pi r}$
$\rho(x)$
$1/\pi r$
0
$-r$
0
r
x